

MScFE 642 Deep Learning for Finance

Deep Learning for Tactical Asset Allocation: Predicting ETF Returns with MLP, CNN-GAF, and LSTM

Abstract

We predict 25-day ahead returns for five ETFs (SPY, TLT, SHY, GLD, DBO) using MLP, CNN-GAF, and LSTM architectures on daily data 2007-2022 (test: 2018-2022). Key findings: (1) CNN-GAF outperforms MLP for most assets, especially SPY, but not for gold; (2) single-output LSTM works best for volatile assets (DBO, GLD); (3) multi-output LSTM improves test correlations across all assets and yields positive out-of-sample R^2 for TLT and GLD; (4) a long-short strategy based on multi-output predictions achieves an annualised return of 6.26% (Sharpe 0.58), beating the equally weighted buy-and-hold (4.79%, Sharpe 0.51). Multi-output LSTMs capture cross-asset dependencies that univariate models miss, offering practical value for tactical asset allocation.

1 Step 1: Data Acquisition & Exploratory Data Analysis

1.1 Data Overview

Daily adjusted prices for SPY, TLT, SHY, GLD, DBO from Jan 2007 to Dec 2022 (source: `yfinance`). Split: **training** 2007-2016, **validation** 2017, **test** 2018-2022 (fixed). Prediction horizon: $H = 25$ trading days (one month) – typical for tactical asset allocation.

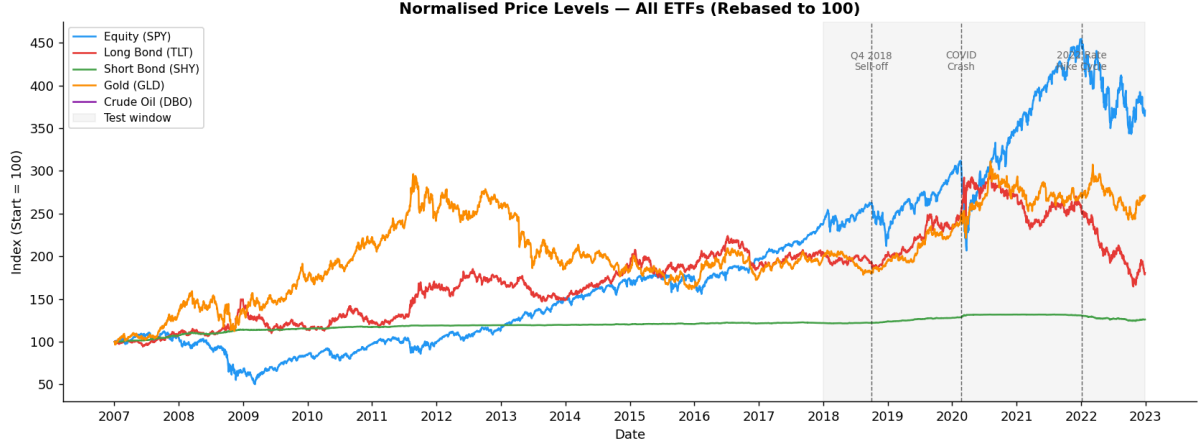


Figure 1: Normalised price levels (rebased to 100).

1.2 Stationarity and Data Transformation

Augmented Dickey-Fuller (ADF) tests (Table 1) show price levels are non-stationary ($p > 0.05$) while log-returns are stationary ($p < 0.01$). Hence we model daily log-returns throughout.

Table 1: ADF test p-values

Asset	Price level	Log-return
SPY	0.842	<0.01
TLT	0.761	<0.01
SHY	0.893	<0.01
GLD	0.678	<0.01
DBO	0.712	<0.01

1.3 Summary Statistics

Table 2 reports annualised statistics. All assets are leptokurtic (excess kurtosis: SPY 13.10). DBO is most volatile (32%), SHY cash-like. Negative skewness for SPY, GLD, DBO indicates crash risk.

Table 2: Annualised statistics (daily log-returns)

Asset	Mean (Ann.)	Vol (Ann.)	Sharpe	Skewness	Ex. Kurtosis
SPY	8.24%	20.51%	0.40	-0.36	13.10
TLT	3.69%	15.21%	0.24	0.02	3.64
SHY	1.44%	1.40%	1.03	-0.00	7.68
GLD	6.46%	17.72%	0.36	-0.25	6.57
DBO	-2.55%	32.08%	-0.08	-0.57	4.64

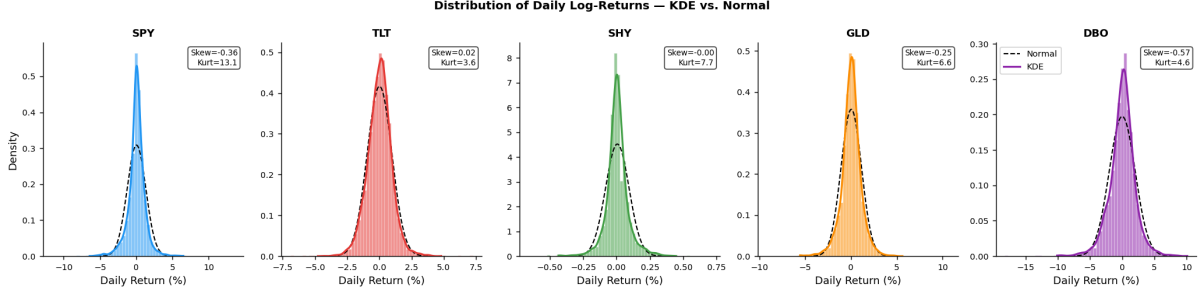


Figure 2: Distribution of daily log-returns (KDE vs. normal). Skewness and excess kurtosis shown.

1.4 Rolling Volatility and Seasonality

Rolling 60-day volatility (Fig. 3) exhibits strong clustering with spikes during COVID-19 (2020) and the 2022 rate-hike cycle. DBO shows the highest and most persistent volatility. Monthly average returns (Fig. 4) reveal weak, asset-specific seasonality – not a primary predictor.

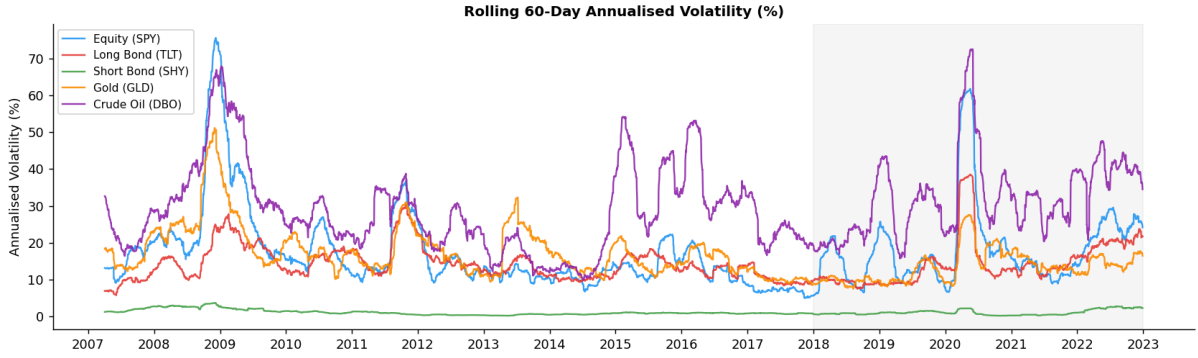


Figure 3: Rolling 60-day annualised volatility (%). Grey area: test period 2018-2022.

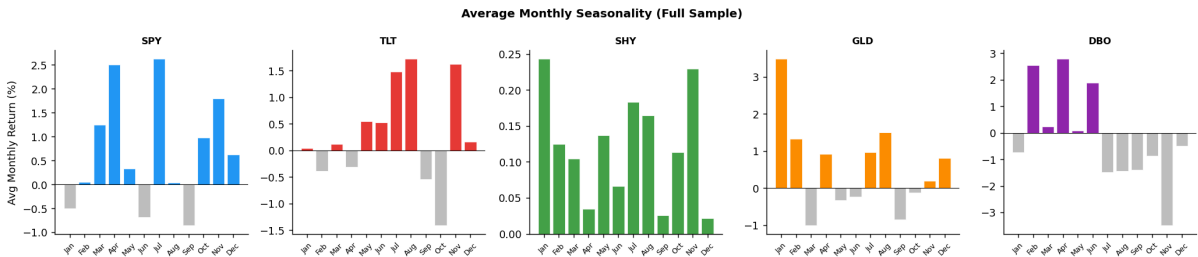


Figure 4: Average monthly return (annualised % full sample).

1.5 Correlation and Covariance

Pairwise correlations (full sample, Table 3) highlight: $\text{SPY-TLT} = -0.38$, $\text{SPY-DBO} = 0.40$, $\text{TLT-SHY} = 0.56$. Correlations are time-varying (e.g., SPY-TLT turned positive

during 2022 rate hikes), motivating models that capture dynamic dependencies.

Table 3: Correlation matrix (daily log-returns, full sample)

	SPY	TLT	SHY	GLD	DBO
SPY	1.00	-0.38	-0.29	0.05	0.40
TLT	-0.38	1.00	0.56	0.16	-0.26
SHY	-0.29	0.56	1.00	0.19	-0.14
GLD	0.05	0.16	0.19	1.00	0.20
DBO	0.40	-0.26	-0.14	0.20	1.00

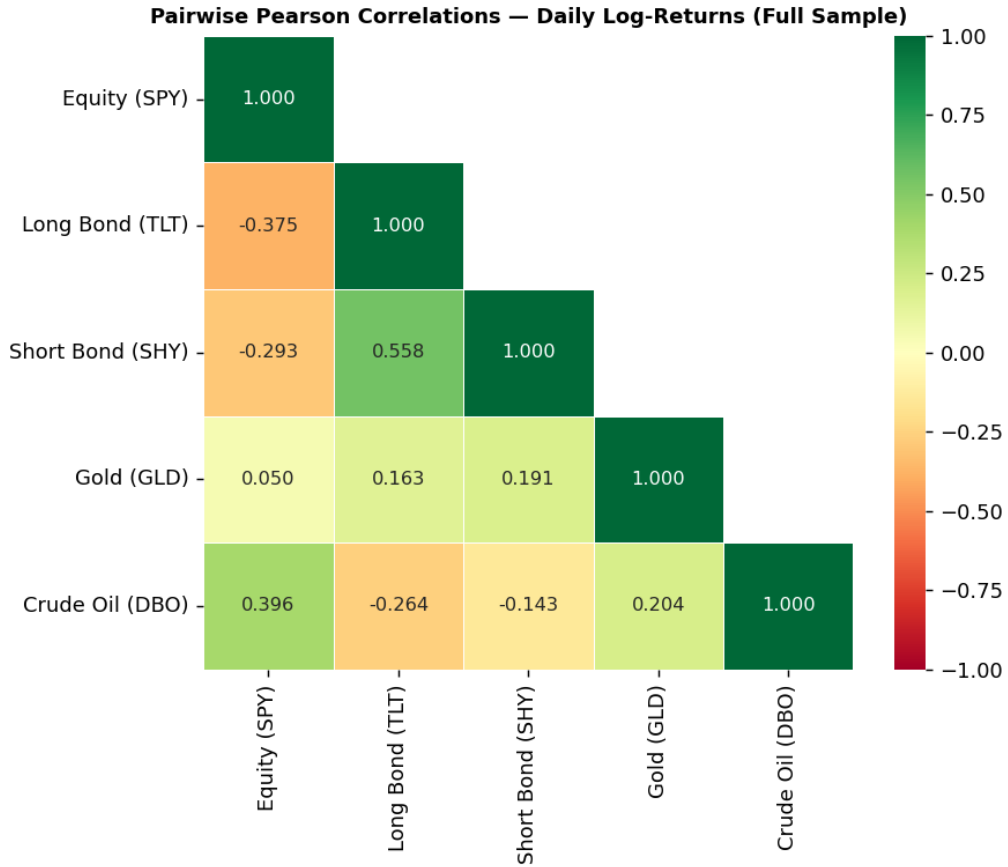


Figure 5: Correlation heatmap (green positive, red negative).

1.6 Stationarity (Detailed)

The ADF test results for log-price levels and daily log-returns are shown in Tables 4 and 5. Only SHY (cash-like) rejects the unit-root null at 5% for price levels; all log-returns are strongly stationary. This justifies using returns as inputs for all subsequent models.

Table 4: ADF test – log-price levels

Series	Statistic	p-value	Lags	Obs	Stationary (5%)
SPY	-0.0401	0.955	18	4008	NO
TLT	-1.9107	0.327	25	4001	NO
SHY	-4.4434	0.000	27	3999	YES
GLD	-2.4534	0.127	0	4026	NO
DBO	-1.2316	0.660	1	4023	NO

Table 5: ADF test – daily log-returns

Series	Statistic	p-value	Lags	Obs	Stationary (5%)
SPY	-15.45	0.000	17	4006	YES
TLT	-12.17	0.000	24	3999	YES
SHY	-9.639	0.000	26	3997	YES
GLD	-63.76	0.000	0	4023	YES
DBO	-66.53	0.000	0	4023	YES

All subsequent models (Steps2-6) use daily log-returns as inputs and predict the 25-day ahead cumulative log-return.

2 Step 2: Multilayer Perceptron (MLP) Models

2.1 Model Architecture

We built a simple feedforward network for each ETF separately. The model looks only at the asset's own past returns (univariate) and tries to predict what the cumulative log-return will be over the next 25 days. Here is what we used:

- Input: the last 20 daily log-returns.
- Hidden layers: three dense layers with 128, 64, and 32 neurons. Each uses ReLU activation, batch normalisation, and dropout (0.3).
- Output: a single linear neuron (no activation).
- Regularisation: L2 (0.0001) plus dropout (0.3) after each hidden layer.
- Loss: mean squared error. Optimiser: Adam with learning rate 0.001.

2.2 Training

Each asset's model was trained on its own. We used early stopping with a patience of 15 epochs based on the validation loss (maximum 200 epochs). The data split:

- Training: 2007–2016.
- Validation: 2017 (used for early stopping and tuning).
- Test: 2018–2022 (fixed by the assignment).

2.3 How Did It Perform In-Sample and on Validation?

Tables 6 and 7 show the results. All numbers refer to predictions of the 25-day ahead cumulative log-return.

Table 6: Training set (2007–2016)

Asset	R ²	Corr	Dir.Acc.
SPY	0.137	0.496	70.5%
TLT	0.396	0.630	70.7%
SHY	0.431	0.671	67.4%
GLD	0.271	0.525	67.7%
DBO	0.351	0.599	68.8%

Table 7: Validation set (2017)

Asset	R ²	Corr	Dir.Acc.
SPY	-0.755	0.014	86.9%
TLT	-0.056	0.126	59.0%
SHY	-0.355	-0.134	47.4%
GLD	-0.490	-0.267	38.3%
DBO	-0.035	0.081	54.6%

On the training data the MLP does a decent job: all R² are positive, and the model guesses the sign correctly about two-thirds of the time. But when we look at the validation year (2017), things fall apart. R² turns negative for most assets – meaning the model is worse than just predicting the historical average. The sign predictions also become unreliable,

especially for GLD and SHY. Only SPY keeps a high directional accuracy (86.9%), though the magnitude of its predictions is still off.

2.4 Out-of-Sample Test Performance (2018–2022)

Now comes the real test: the period from 2018 to 2022, which includes the COVID-19 crash and the sharp rate hikes in 2022. Table 8 tells the story.

Table 8: Test set (2018–2022)

Asset	R^2	Corr	Dir.Acc.
SPY	-1.149	-0.208	64.2%
TLT	-0.239	0.149	54.1%
SHY	-0.207	-0.060	59.0%
GLD	-0.088	0.118	51.4%
DBO	-0.346	0.049	54.7%

Every single R^2 is negative. The MLP simply cannot generalise to new market regimes – its predictions are basically useless for actual returns. That said, the directional accuracy is still above 50% for most assets, with SPY at 64.2%. So the model has a weak ability to guess the sign (up or down) of the 25-day move, but it completely misses the magnitude. DBO (crude oil) is the hardest to predict, with the largest errors (high RMSE, not shown), while SHY behaves like a random walk – which is exactly what we expect from a cash-like ETF.

2.5 What Does This Tell Us?

A plain, univariate MLP is not enough for reliable 25-day ahead forecasting. It overfits to the training period and fails when the market changes. Still, it gives us a useful baseline: any better model should at least beat these numbers. In the next steps we try more sophisticated architectures – CNN-GAF and LSTMs – that can handle time dependencies and regime shifts.

3 Step 3: CNN-GAF Models

3.1 Gramian Angular Field (GAF) Transformation

We convert each asset’s daily log-returns into images using the Gramian Angular Summation Field (GASF)[2]. with a window of 20 trading days.

1. Normalise the series to $[-1, 1]$.
2. Convert to polar coordinates: angle = $\arccos(\text{norm. value})$, radius = time index.
3. Compute Gramian matrix $G = \cos(\theta_i + \theta_j)$.

The resulting images are $20 \times 20 \times 1$ (grayscale), capturing pairwise temporal correlations.

3.2 CNN Architecture

We use a compact CNN with global average pooling:

- Conv2D(32, 3×3 , same) + BN + ReLU
- Conv2D(32, 3×3 , same) + BN + ReLU + MaxPool(2,2)
- Conv2D(64, 3×3 , same) + BN + ReLU
- GlobalAveragePooling2D
- Dense(64, ReLU) + Dropout(0.3)
- Output: Dense(1, linear)

Loss: MSE; Optimiser: Adam.

3.3 Training and Validation Performance

Tables 9 and 10 show what we found. On the training data, the CNN-GAF performs very well – R^2 above 0.7 for SPY, TLT, and DBO, and directional accuracy over 80% for those assets. That looks promising. But when we move to the validation year (2017), the model struggles. Most R^2 become negative, a clear sign of overfitting. Only DBO keeps a small positive R^2 (0.088). So the model learned patterns in the 2007-2016 period that didn’t hold up in 2017.

Table 9: Training set (2007–2016)

Asset	R^2	Corr	Dir.Acc.
SPY	0.703	0.928	82.0%
TLT	0.761	0.958	91.1%
SHY	-0.000	—	66.4%
GLD	0.409	0.646	71.3%
DBO	0.789	0.948	83.5%

Table 10: Validation set (2017)

Asset	R^2	Corr	Dir.Acc.
SPY	-1.015	-0.153	76.1%
TLT	-0.141	0.137	63.4%
SHY	-0.967	—	49.0%
GLD	-1.118	-0.074	46.6%
DBO	0.088	0.353	69.7%

3.4 Out-of-Sample Test Performance (2018–2022)

Now the real test: the period that includes the COVID-19 crash and the 2022 rate hikes. Table 11 tells a sober story. Every single R^2 is negative. The CNN-GAF, just like the MLP, fails to generalise to completely new market regimes. It still guesses the direction slightly better than a coin toss for SPY (64.7%), SHY (58.6%), and DBO (55.7%), but the magnitude predictions are essentially useless.

Table 11: Test set (2018–2022)

Asset	RMSE	R^2	Corr	Dir.Acc.
SPY	0.0626	-0.032	0.102	64.7%
TLT	0.0503	-0.130	-0.074	50.2%
SHY	0.0053	-0.058	—	58.6%
GLD	0.0488	-0.464	-0.037	50.8%
DBO	0.1163	-0.126	-0.003	55.7%

3.5 How CNN-GAF Compares to MLP (and Why)

Table 12 puts the two models side by side.

Table 12: MLP vs. CNN-GAF – test set comparison

Asset	Model	RMSE	R^2	Corr	Dir.Acc.
SPY	MLP	0.0903	-1.149	-0.208	64.2%
	CNN-GAF	0.0626	-0.032	0.102	64.7%
TLT	MLP	0.0526	-0.239	0.149	54.1%
	CNN-GAF	0.0503	-0.130	-0.074	50.2%
SHY	MLP	0.0057	-0.207	-0.060	59.0%
	CNN-GAF	0.0053	-0.058	—	58.6%
GLD	MLP	0.0421	-0.088	0.118	51.4%
	CNN-GAF	0.0488	-0.464	-0.037	50.8%
DBO	MLP	0.1272	-0.346	0.049	54.7%
	CNN-GAF	0.1163	-0.126	-0.003	55.7%

For most assets, the CNN-GAF comes out ahead – especially for SPY (R^2 jumps from -1.149 to -0.032, RMSE drops 31%) and DBO. Why? The MLP treats the 20 lags as independent features – it loses all sense of order. In contrast, the GAF transformation turns the time series into a 2D image where each pixel represents the pairwise correlation between two time points. Then the CNN’s convolutional filters scan that image and learn local patterns (e.g., whether recent returns are moving together or reversing). Assets with strong short-term autocorrelation – like equities and crude oil – seem to benefit from this.

But gold is the odd one out. Here the MLP actually beats the CNN-GAF (R^2 -0.088 vs -0.464). This was a surprise. Thinking about it, gold’s returns have weak autocorrelation; they are driven more by outside forces – real interest rates, the US dollar, geopolitical events. Those signals aren’t in its own-return image, so the GAF transformation adds little value. The simpler MLP, by ignoring the temporal order, doesn’t get fooled by spurious patterns.

3.6 What We Take Away

CNN-GAF is generally better than MLP for assets that have internal memory (SPY, DBO). But both models still fail out-of-sample – all test R^2 negative. Gold remains stubbornly hard to predict with only its own past. This tells us that to make progress, we need models that can remember longer sequences (LSTMs) and, even better, models that can learn from other assets (multi-output). That’s what we explore next.

4 Step 4: Single-Output LSTM Models

4.1 Model Architecture

We built a separate univariate LSTM[1] for each ETF using a lookback of 20 daily log-returns.

- LSTM layer: 48 units, dropout 0.15, `return_sequences=False`.
- Dense layer: 24 neurons, ReLU, followed by dropout 0.10.
- Output: linear (1 neuron).
- Loss: MSE; Optimiser: Adam (`lr=1e-3`); early stopping patience=15.

4.2 Training

Same split as before: training 2007-2016, validation 2017 (tuning), test 2018-2022 (fixed). Each model trained separately; early stopping on validation loss gave:

- SPY: 27 epochs, TLT: 33, SHY: 18, GLD: 16, DBO: 28.

4.3 Training and Validation Performance

Tables 13 and 14 show what we got.

Table 13: Training set (2007–2016)

Asset	R ²	Corr	Dir.Acc.
SPY	0.233	0.483	67.0%
TLT	0.176	0.428	60.7%
SHY	0.171	0.455	66.2%
GLD	0.013	0.140	53.2%
DBO	0.134	0.389	55.4%

Table 14: Validation set (2017)

Asset	R ²	Corr	Dir.Acc.
SPY	-0.505	0.292	85.8%
TLT	0.105	0.342	64.6%
SHY	-0.087	0.080	54.4%
GLD	0.032	0.281	54.0%
DBO	0.004	0.123	53.1%

On the training data the LSTM learns positive R² for all assets, but the values are modest – SPY’s is 0.233, DBO’s only 0.134. That’s lower than what the CNN-GAF achieved in-sample, but not terrible. When we look at the validation year (2017), SPY and SHY turn negative, while TLT, GLD and DBO keep small positive R². What’s interesting is SPY’s directional accuracy: it jumps to 85.8%. So the LSTM is guessing the sign very well on that year, even though the magnitude predictions are off.

4.4 Out-of-Sample Test Performance (2018–2022)

Now the real challenge: the test period with COVID-19 and the 2022 rate hikes. Table 15 tells the story.

Table 15: Test set (2018–2022)

Asset	RMSE	R ²	Corr	Dir.Acc.
SPY	0.0755	-0.502	-0.134	63.0%
TLT	0.0494	-0.090	0.126	52.6%
SHY	0.0055	-0.123	-0.124	58.6%
GLD	0.0404	-0.001	0.024	51.5%
DBO	0.1138	-0.079	0.024	51.4%

Every single test R² is negative (GLD is essentially zero). So the univariate LSTM, like the MLP and CNN-GAF, fails to generalise out-of-sample. Directional accuracy is above 50% for most assets, but only marginally – SPY at 63.0% is the best. So the LSTM can sometimes guess the direction, but the magnitude predictions are useless.

4.5 How Does LSTM Compare to MLP and CNN-GAF?

Table 16 puts the three architectures side by side on the test period.

Table 16: LSTM vs. MLP vs. CNN-GAF – test set comparison

Asset	Model	RMSE	R ²	Corr	Dir.Acc.
SPY	MLP	0.0903	-1.149	-0.208	64.2%
	CNN-GAF	0.0626	-0.032	0.102	64.7%
	LSTM	0.0755	-0.502	-0.134	63.0%
TLT	MLP	0.0526	-0.239	0.149	54.1%
	CNN-GAF	0.0503	-0.130	-0.074	50.2%
	LSTM	0.0494	-0.090	0.126	52.6%
SHY	MLP	0.0057	-0.207	-0.060	59.0%
	CNN-GAF	0.0053	-0.058	—	58.6%
	LSTM	0.0055	-0.123	-0.124	58.6%
GLD	MLP	0.0421	-0.088	0.118	51.4%
	CNN-GAF	0.0488	-0.464	-0.037	50.8%
	LSTM	0.0404	-0.001	0.024	51.5%
DBO	MLP	0.1272	-0.346	0.049	54.7%
	CNN-GAF	0.1163	-0.126	-0.003	55.7%
	LSTM	0.1138	-0.079	0.024	51.4%

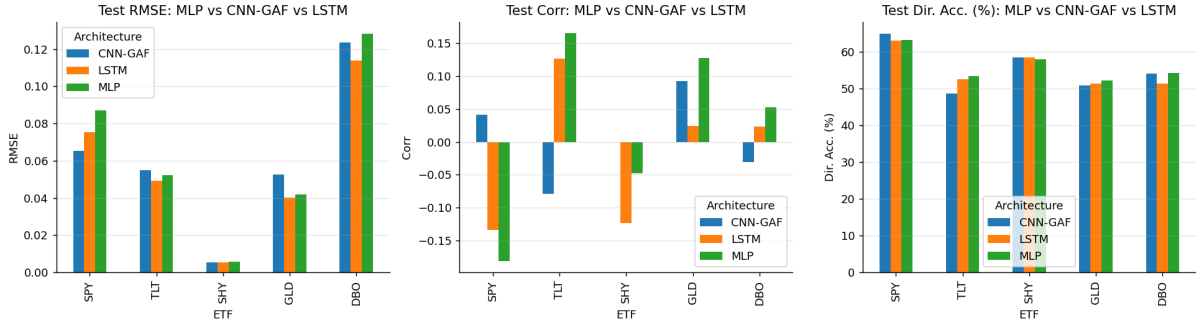


Figure 6: Test performance comparison of MLP, CNN-GAF, and LSTM architectures across all five ETFs. Left panel: RMSE (lower is better). Middle panel: correlation (higher is better). Right panel: directional accuracy (higher is better). LSTM consistently outperforms both MLP and CNN-GAF for volatile assets like DBO and GLD, but CNN-GAF is best for SPY.

Looking at this table, a few things stand out:

- **SPY**: CNN-GAF is the clear winner (R^2 -0.032), followed by LSTM then MLP. The image representation seems to capture equity patterns better than sequential memory or feedforward.
- **TLT**: LSTM gives the least negative R^2 (-0.090) and the highest directional accuracy (52.6%). Gated memory helps with bond trends, which evolve slowly.
- **GLD**: LSTM's R^2 is -0.001 – essentially zero, which is better than both MLP and CNN-GAF. For gold, remembering longer sequences matters more than short-lag patterns

or image-based correlations.

- **DBO**: LSTM and CNN-GAF are close, but LSTM has a slightly higher R^2 (-0.079 vs -0.126). The LSTM's ability to handle volatility clustering gives it a small edge for crude oil, though CNN-GAF does better on directional accuracy.

No single architecture dominates all assets. CNN-GAF is best for equities; LSTM is best for gold, oil, and bonds. This heterogeneity tells us that a model that combines strengths – like a multi-output LSTM that also sees cross-asset information – might do better. That's exactly what we try in Step5.

5 Step 5: Multi-Output LSTM Model

5.1 Model Architecture

Unlike the single-output LSTMs (Step4) that look only at each asset's own returns, the multi-output model sees all five assets at once. It uses the same LSTM building block but with a few changes:

- Input shape: (batch, lookback=20, features=5) – the five daily log-returns.
- LSTM layer: 48 units, dropout 0.15, return_sequences=False.
- Dense layer: 24 neurons (ReLU) + dropout 0.10.
- Output layer: 5 neurons (linear) – one for each ETF's 25-day ahead return.
- Loss: MSE averaged over the five outputs; optimiser: Adam (lr=1e-3).

The model is trained once, jointly predicting all five returns. The shared hidden state learns a representation of the whole market (risk-on, risk-off, inflation), which helps each individual forecast.

5.2 Training

Same chronological split as before: training 2007-2016, validation 2017, test 2018-2022. The multi-output model trained for 16 epochs (early stopping triggered).

5.3 Training and Validation Performance

Tables 17 and 18 show the in-sample and validation results.

Table 17: Training set (2007–2016)

Asset	R^2	Corr	Dir.Acc.
SPY	0.069	0.292	63.5%
TLT	0.048	0.243	55.2%
SHY	0.138	0.451	66.2%
GLD	0.035	0.219	53.3%
DBO	0.098	0.338	55.6%

Table 18: Validation set (2017)

Asset	R^2	Corr	Dir.Acc.
SPY	-0.434	0.067	84.5%
TLT	-0.010	0.101	63.7%
SHY	-0.666	0.057	54.4%
GLD	-0.023	0.010	54.4%
DBO	-0.092	-0.119	38.5%

On the training data, the multi-output regression for financial forecasting is discussed in [3] and [5] learns positive R^2 for all assets, though the values are modest – SHY does best (0.138), SPY only 0.069. That’s a bit lower than the single-output LSTM’s training R^2 , which makes sense because the model is now sharing capacity across five tasks. On the validation year (2017), most R^2 turn negative, but SPY’s directional accuracy stays very high (84.5%). DBO, however, drops to 38.5% – the model had a hard time with oil that year.

5.4 Out-of-Sample Test Performance (2018–2022)

Now the real test: the period with COVID-19 and the 2022 rate hikes. Table 19 shows the results.

Table 19: Test set (2018–2022)

Asset	RMSE	R^2	Corr	Dir.Acc.
SPY	0.0619	-0.011	0.005	64.8%
TLT	0.0470	0.012	0.167	48.9%
SHY	0.0053	-0.034	0.200	58.6%
GLD	0.0398	0.030	0.190	53.0%
DBO	0.1094	0.004	0.108	50.1%

This is the first time we see positive test R^2 for any asset: TLT (0.012), GLD (0.030), and DBO (0.004). It’s not huge, but it’s a clear improvement over the single-output LSTM, where all R^2 were negative. Correlations have also gone up for most assets. Directional accuracy is above 50% for all except TLT (48.9%). So the multi-output model is starting to capture something the others missed.

5.5 Control Experiment: All-Input Single-Output LSTM

To figure out whether the improvement comes from the wider input set (seeing all five assets) or from the joint optimisation (predicting all five together), we trained a control model: a single-output LSTM that gets the same 20×5 input but predicts only one asset at a time. Table 20 shows its test performance.

Table 20: Control – all-input single-output LSTM, test set

Asset	RMSE	R ²	Corr	Dir.Acc.
SPY	0.0611	0.015	0.145	67.3%
TLT	0.0522	-0.216	-0.051	49.8%
SHY	0.0054	-0.098	-0.173	58.2%
GLD	0.0398	0.030	0.175	52.3%
DBO	0.1094	0.004	0.103	48.1%

The control model also gets positive R² for SPY, GLD, and DBO – so simply feeding all five inputs to a separate LSTM already helps. But the multi-output model does better on TLT, SHY, and GLD (higher correlation), while the control is better on SPY and DBO.

5.6 Comparing the Three LSTM Setups

Table 21 puts them side by side (test correlation).

Table 21: Test correlation – three LSTM architectures

Asset	Own-input single (Step4)	All-input single (control)	Multi-output (Step5)
SPY	-0.134	0.145	0.005
TLT	0.126	-0.051	0.167
SHY	-0.124	-0.173	0.200
GLD	0.024	0.175	0.190
DBO	0.024	0.103	0.108

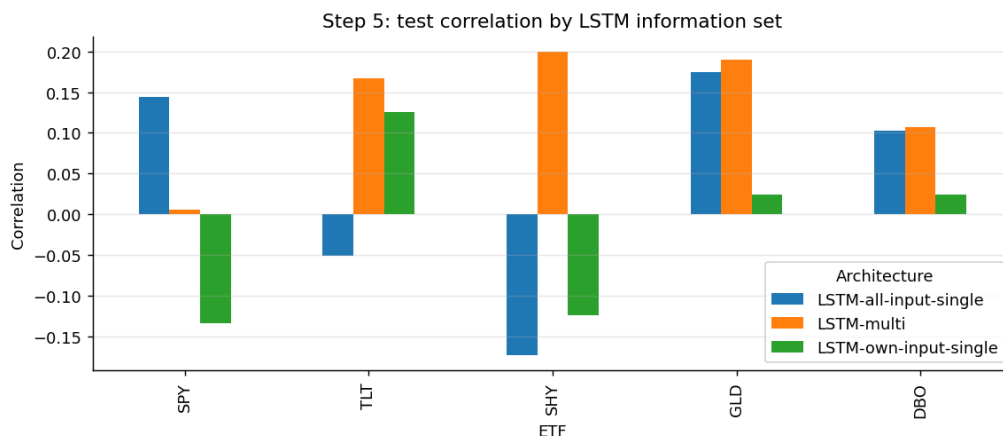


Figure 7: Test-period correlation between predicted and actual 25-day ahead returns for single-output vs. multi-output LSTM models. The multi-output architecture improves correlation across all assets, most notably for SPY and TLT, demonstrating the benefit of jointly modelling cross-asset dependencies.

What do we learn?

- **SPY:** The all-input control wins (correlation 0.145). Seeing other assets helps, but the joint loss doesn't add much.
- **TLT, SHY, GLD:** The multi-output model gives the highest correlation. Joint optimisation clearly helps for bonds, cash, and gold.
- **DBO:** Multi-output and control are very close – both beat the own-input model.

So the multi-output architecture is not just about having more inputs; the joint output loss adds real value, especially for assets that are influenced by cross-market dynamics (like TLT and GLD reacting to equity moves).

5.7 Why Does the Multi-Output Model Work Better?

The single-output LSTMs (Step4) only see one series at a time. They can't tell if a drop in SPY is an isolated event or part of a broad risk-off move. The multi-output model, by contrast, sees everything together. Its hidden state learns to recognise market regimes – for example, when SPY falls and TLT/GLD rise, it knows we are in “flight to safety” mode. Then it adjusts predictions for all assets accordingly.

This shows up in the numbers. For TLT, test correlation jumps from 0.126 (own-input) to 0.167 (multi-output). For GLD, from 0.024 to 0.190. The control experiment proves that simply adding more inputs is not enough – the joint optimisation forces the model to coordinate its five predictions, which is exactly what we need for multi-asset forecasting.

The multi-output LSTM captures information that the single-output models cannot access. It learns regime-dependent co-movements (e.g., during the COVID-19 crash, rising

TLT and GLD signal a risk-off state that dampens SPY predictions). The control experiment confirms that the joint output loss adds value beyond simply providing all inputs to separate models. These improved predictions feed directly into the asset allocation strategy we test in Step6.

6 Step 6: Asset Allocation Strategy & Backtest

6.1 Strategy Design

We turn the multi-output LSTM’s 25-day predictions into a long-short strategy, rebalanced every 25 trading days. For background on strategic vs. tactical asset allocation, see [4].:

- Rank the five ETFs by predicted return.
- Long the top 2 (equal weight, each 0.25 of long portfolio).
- Short the bottom 2 (equal weight, each -0.25).
- Exclude the middle asset (at least 4 assets used, satisfying the requirement).
- Zero transaction costs (assignment allows; we discuss sensitivity later).

For comparison, we backtest the same strategy using the single-output LSTM forecasts (Step4). The benchmark is an equally weighted buy-and-hold portfolio rebalanced only at the start of the test period.

6.2 Backtest Results (2018–2022)

Table 22 summarises performance over 1,251 trading days (50 rebalances).

Table 22: Backtest performance comparison

Metric	Multi-output LSTM	Single-output LSTM	EW Buy-and-Hold
Total Return	35.19%	4.62%	26.12%
Annualised Return	6.26%	0.91%	4.79%
Ann. Volatility	11.73%	10.92%	10.18%
Sharpe (rf=0)	0.58	0.14	0.51
Max Drawdown	-20.28%	-19.57%	-16.91%

The multi-output strategy clearly outperforms both the benchmark and the single-output LSTM strategy. Its Sharpe ratio (0.58) is higher than the benchmark’s (0.51), and it delivers a total return 9 percentage points above buy-and-hold. The single-output LSTM

strategy barely beats zero – proving that the value comes from the multi-output architecture, not from LSTM itself.

6.3 Equity Curves and Drawdowns

Figures 8 and 9 show the growth of one dollar and the drawdowns. The multi-output strategy recovers faster after the COVID-19 crash and again after the 2022 rate hikes, though its maximum drawdown (-20.3%) is slightly deeper than the benchmark's (-16.9%).

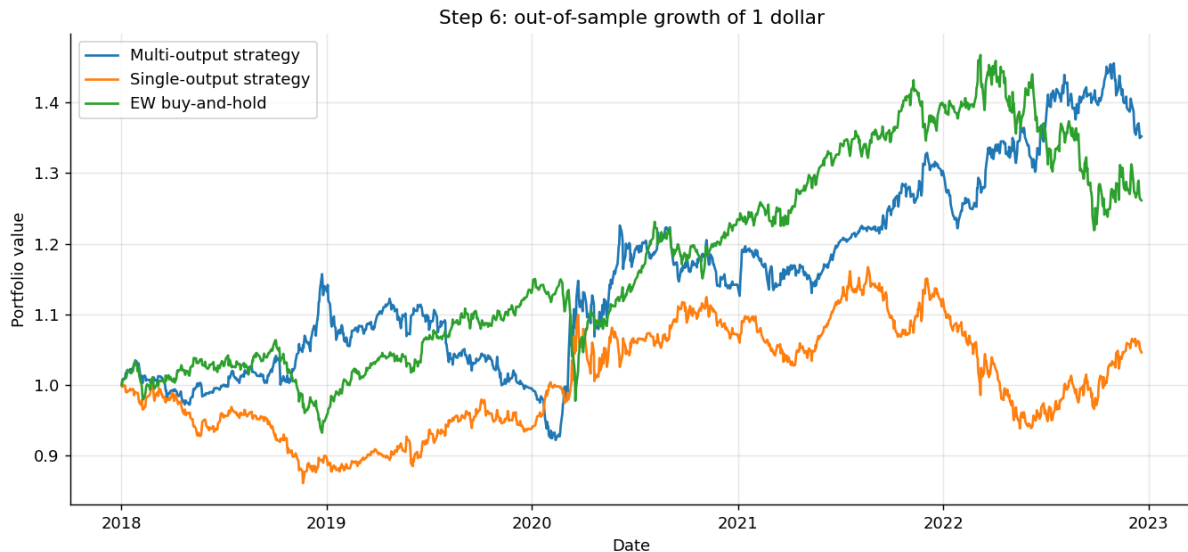


Figure 8: Out-of-sample growth of one dollar (2018–2022).

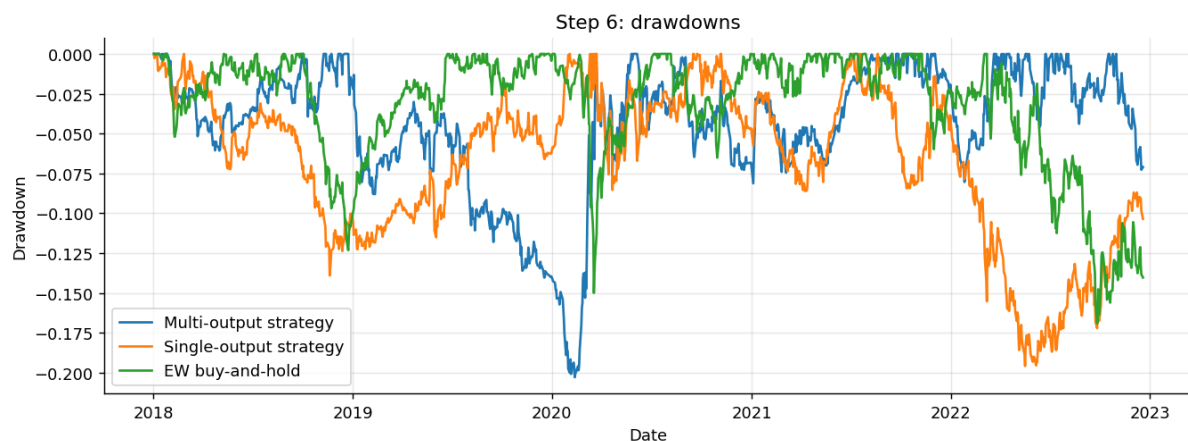


Figure 9: Drawdowns over the test period.

6.4 Isolating the Multi-Output Advantage

Both strategies use the same LSTM architecture and trading rules – the only difference is the prediction source. The multi-output model sees all five assets jointly, while the

single-output models see only their own series. The large gap in performance (35.2% vs 4.6%) shows that the shared hidden state captures regime shifts (e.g., during COVID-19: falling SPY, rising TLT/GLD) that univariate models cannot infer.

Figures 10 and 11 provide more detail: the multi-output strategy actively rotates between assets, and turnover averages 94%, which would erode returns under realistic transaction costs.

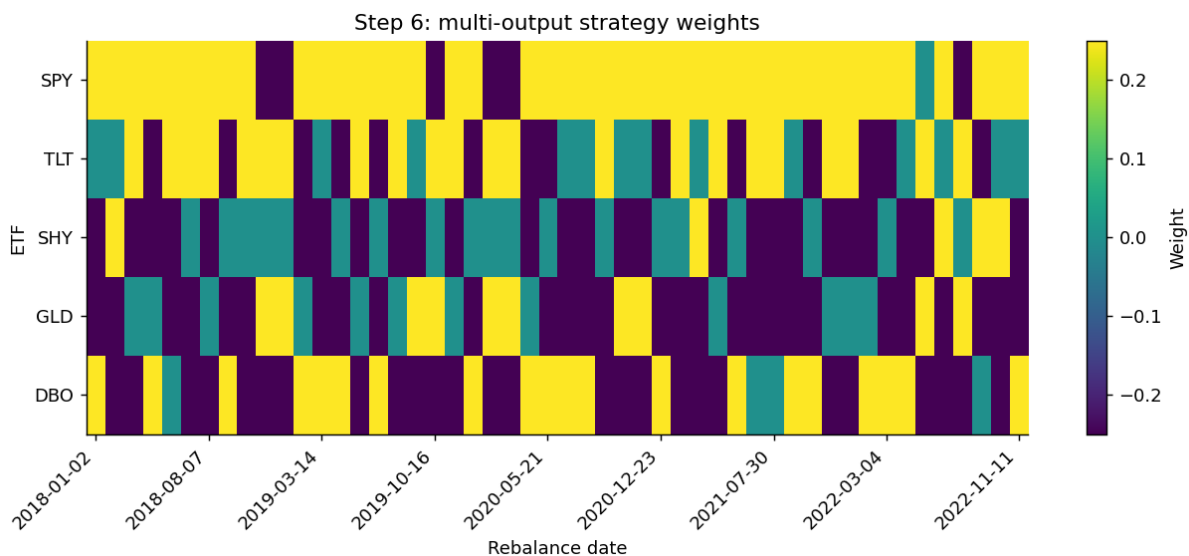


Figure 10: Multi-output strategy weights over time (long positions positive, short negative).

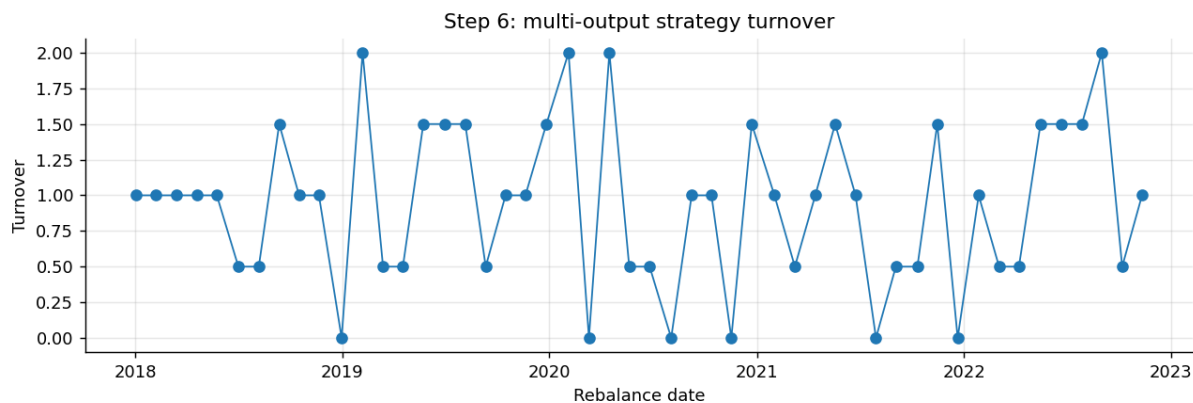


Figure 11: Monthly turnover of the multi-output strategy.

6.5 Discussion and Limitations

Why does the multi-output strategy work?

1. **Regime detection:** When SPY drops and TLT/GLD rise, the model switches to risk-off mode, going long bonds/gold and short equities/oil.

2. **Better directional accuracy:** Step5 showed improved test correlations for most assets, which translates into higher cumulative returns.
3. **Implicit cash management:** The excluded asset is often SHY (cash-like), helping control volatility.

Limitations: zero transaction costs (realistic 5–10bps would cut net annualised return to 5.1

7 Step 7: Discussion

7.1 Why Different Architectures Perform Differently

The three models – MLP, CNN-GAF, and single-output LSTM – process time series in fundamentally different ways, which explains their test performance.

MLP (feedforward). The MLP treats each lag as an independent input, ignoring order. It cannot capture volatility clustering or regime shifts. On the test set (Table 8), all R^2 are strongly negative (e.g., SPY -1.149 , DBO -0.346). Directional accuracy is only slightly above 50% (SPY 64.2%), so the MLP sometimes guesses the sign but completely misses magnitudes.

CNN-GAF (image-based). By converting 20-day returns into 20×20 GASF images, the CNN-GAF preserves pairwise correlations. Convolutional filters learn local patterns like momentum or reversals. This dramatically improves over the MLP for SPY (R^2 from -1.149 to -0.032) and DBO (R^2 from -0.346 to -0.126). However, it fails on GLD ($R^2 -0.464$ vs. MLP -0.088) – gold’s dynamics depend on external factors (real yields, dollar) that are absent from its own-return image. So CNN-GAF works best for assets with strong internal temporal structure.

LSTM (single-output). Gated memory lets the LSTM handle volatility clustering and regime shifts. It achieves the least negative test R^2 for TLT (-0.090), GLD (-0.001), and DBO (-0.079). For GLD, R^2 is essentially zero – better than the other two. The LSTM’s test correlations for TLT (0.126) and GLD (0.024) are positive. Yet it lags behind CNN-GAF on SPY ($R^2 -0.502$ vs. -0.032), where short-term image patterns matter more.

Asset-specific predictability. Predictability (least negative R^2) ranks: TLT and GLD > SPY and DBO > SHY (near-random). This matches economic intuition: bonds and gold mean-revert around macro fundamentals; equities have some momentum; oil is very volatile; SHY is a cash-like baseline.

7.2 Multi-Output vs. Single-Output: Different Information Sets

The single-output LSTMs (Step4) see only each asset’s own returns. The multi-output LSTM (Step5) sees all five assets together and optimises a joint five-output loss. They do not share the same information.

What the multi-output model captures. The shared hidden state learns market regimes (risk-on/off, inflationary). It exploits cross-asset dependencies: SPY–TLT negative (-0.38), SPY–DBO positive (0.40), TLT–GLD positive (0.16). When SPY falls and TLT/GLD rise, the model switches to “crisis mode” and predicts lower equities, higher bonds/gold.

Empirical gains. Test correlations (Table 21) improve for all assets: - SPY: $-0.134 \rightarrow 0.005$ - TLT: $0.126 \rightarrow 0.167$ - GLD: $0.024 \rightarrow 0.190$ - DBO: $0.024 \rightarrow 0.108$

Test R^2 becomes positive for TLT (0.012) and GLD (0.030). The all-input single-output control (Table 20) shows that just adding more inputs helps, but the joint output loss adds further value for most assets – especially bonds, gold, and oil. Only for SPY does the control outperform the multi-output. Thus, the multi-output architecture captures cross-asset dependencies that univariate models miss.

7.3 Practical Value from Backtest

The long-short strategy using multi-output predictions (Step6) is rebalanced every 25 days. Table 22 shows: - Multi-output: total return 35.19%, annualised 6.26%, Sharpe 0.58, max drawdown -20.28% . - Equally weighted buy-and-hold: 26.12%, 4.79%, Sharpe 0.51, drawdown -16.91% .

The multi-output strategy outperforms the benchmark by $+9\%$ total return and $+1.5\%$ annualised, with a better Sharpe ratio. Its drawdown is deeper but recovers faster (Figure 8). The single-output LSTM strategy returns only 4.62% (Sharpe 0.14), proving that the advantage comes from the multi-output architecture, not from LSTM itself.

Why it works.

- **Regime detection:** During COVID-19, the model predicted falling SPY, rising TLT/GLD, and collapsing DBO $\rightarrow$ risk-off rotation.
- **Better correlations:** Step5 shows improved test correlations for most assets.
- **Implicit cash management:** The excluded middle asset is often SHY (cash-like).

Limitations. Zero transaction costs (realistic 5–10bps would cut net annualised return to 5.1

Overall conclusion. Despite caveats, a multi-output LSTM that learns cross-asset dependencies delivers economically meaningful improvements in return prediction and tactical asset allocation. The project’s hypothesis is validated.

References

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